INCH-POUND

MS51492A
20 October 2011
SUPERSEDING
MS51492
Amendment 1
5 October 1992

DETAIL SPECIFICATION SHEET

# SCREW, MACHNE, PAN HEAD, CROSS-RECESSED, CARBON STEEL, UNF-2A

Inactive for new design after 18 November 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and FF-S-92.

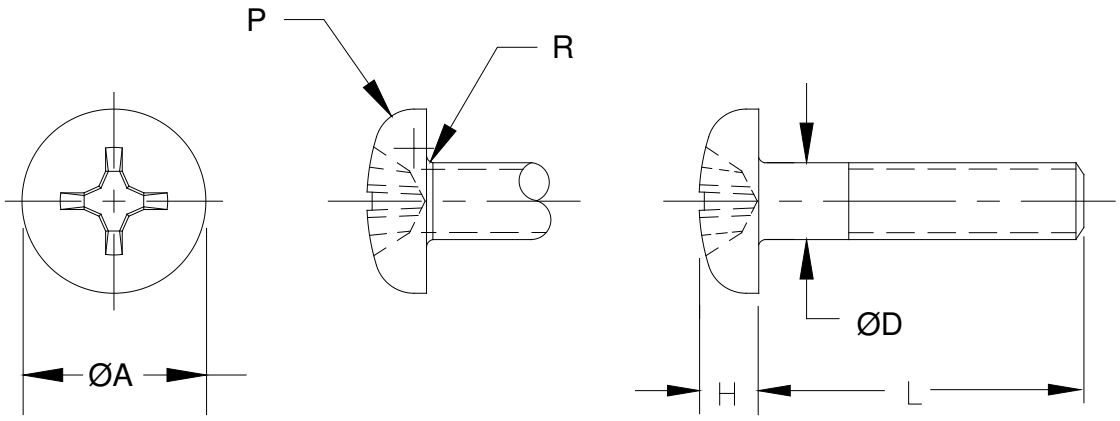



FIGURE 1. SCREW.



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TABLE I. Screw Dash Numbers and Dimensions

| NOMINAL SIZE | | | .112 | .138 | .164 | .190 | .250 |
|---|---|---|---|---|---|---|---|
| THREADS PER INCH UNC-2A | | | 40 | 32 | 32 | 24 | 20 |
| Ø D - BODY DIAMETER | | MAX | .1120 | .1380 | .1640 | .1900 | .2500 |
| | | MIN | .0925 | .1141 | .1399 | .1586 | .2127 |
| Ø A - HEAD DIAMETER | | MAX | .2190 | .2700 | .3220 | .3730 | .4920 |
| | | MIN | .2050 | .2560 | .3060 | .3570 | .4730 |
| H – HEAD HEIGHT | | MAX | .0800 | .0970 | .1150 | .1330 | .1750 |
| | | MIN | .0700 | .0870 | .1050 | .1220 | .1620 |
| P - HEAD RADIUS | | REF | .0100 | .0150 | .0150 | .0200 | .0350 |
| R - FILLET RADIUS | | MAX | .0170 | .0210 | .0250 | .0290 | .0380 |
| MINIMUM TENSILE LOAD – LBS. | | | 360 | 545 | 840 | 1,050 | 1,900 |
| L – LENGTH | TOLERANCE | | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. |
| .250<br>.375<br>.500<br>.625<br>.750<br>1.000 | +.000<br>- .031 | | 01 | 02<br>03<br>04 | 05<br>06<br>07<br>08<br>09<br>10 | 13<br>14<br>15<br>16<br>17 | 19<br>20<br>21<br>22<br>23 |
| 1.250<br>1.500<br>2.000 | +.000<br>- .062 | | | | 11<br>12 | 18 | 24<br>25 |
| 3.000 | +.000<br>- .094 | | | | | | 26 |

NOTES:

1. Material: Carbon steel, in accordance with ASTM-A29/A29M grade 1015 (UNS G10150) thru grade 1030 (UNS G10300) or ASTM A29 grade 1116 (UNS G11160) thru grade 1132 (UNS G11320). The minimum ultimate tensile strength shall be 60,000 PSI.

2. Finish. Zinc coating in accordance with ASTM B633, Fe/Zn5 (5 µm), Type II.

3. Threads: The threads shall be in accordance with FED-STD-H28/2. Acceptability of screw threads shall be in accordance with FED-STD-H 28/20, system 21.

4. Thread length: Screws 2.000 inches long and shorter shall have complete threads extending to within 2 threads of the bearing surface of the head, or closer if practicable. Longer screws have a minimum thread length of 1.750 inches. Threads shall not extend to the fillet radius nor shall threads extend to and undercut the head of the screw.

5. Recess. The recess shall be in accordance with NASM9006.

6. Part number: The basic MS part number is followed by a dash number from Table I.

   Example: MS51492-01 - Screw, Machine, Pan Head, Cross–Recessed, Carbon Steel, .112 - 40UNC-2A, Length .625, Zinc Coated.

7. All dimensions are in inches.



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8. Referenced documents shall be of the issue in effect on date of invitations for bids or request for proposal, except that referenced adopted industry documents shall give the date of the issue adopted.

9. For design feature purposes, this standard takes precedence over documents referenced herein.

10. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AR
Air Force – 99

Preparing activity:
DLA – IS

(Project 5305-2011-009)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil



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